

Embedded projects 2

Weekly exercises 6

Vihtori Virta
Sonja Viinikainen

Exercise 1 - UART

Arduino Mega has multiple UART/[Serial](#) buses, currently most of you are using the default bus to transmit data with the ESP.

- Change your wiring and code to use one of the alternative Serial buses on the board for communicating with the ESP.
 - Consider what data is being sent to which bus and where it is being read from. This will help you ensure that your communication setup is correct.
 - You don't need to change the code or wiring on the ESP side of your system, only the Arduino side.
 - [Example](#) on how to setup multiple serial busses
- Ensure that your new setup is functional and that communication with the ESP is successful.

Exercise 2 – Testing two-way communication

Try out the example code in Moodle → ESP-LIDAR

- Test how fast you can send Lidar values to the webpage
 - Start by sending one Lidar value per second from Arduino Mega and check if the webpage can keep up. Gradually increase the frequency of the values sent.
 - Make a note of the maximum frequency and tell it to us when we check you exercises

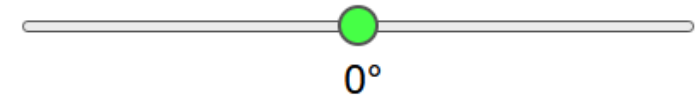
Exercise 3

- Change your Arduino Mega's code to send also your compass data to the ESP and display it on your webpage.
- For now, the data can be displayed just as a numerical value.

Motor Control



Compass Control



Lidar Data

0 cm

Compass value: 45